

ELEC 221 2026W1 Tutorial 1

Monday 14 September 2026

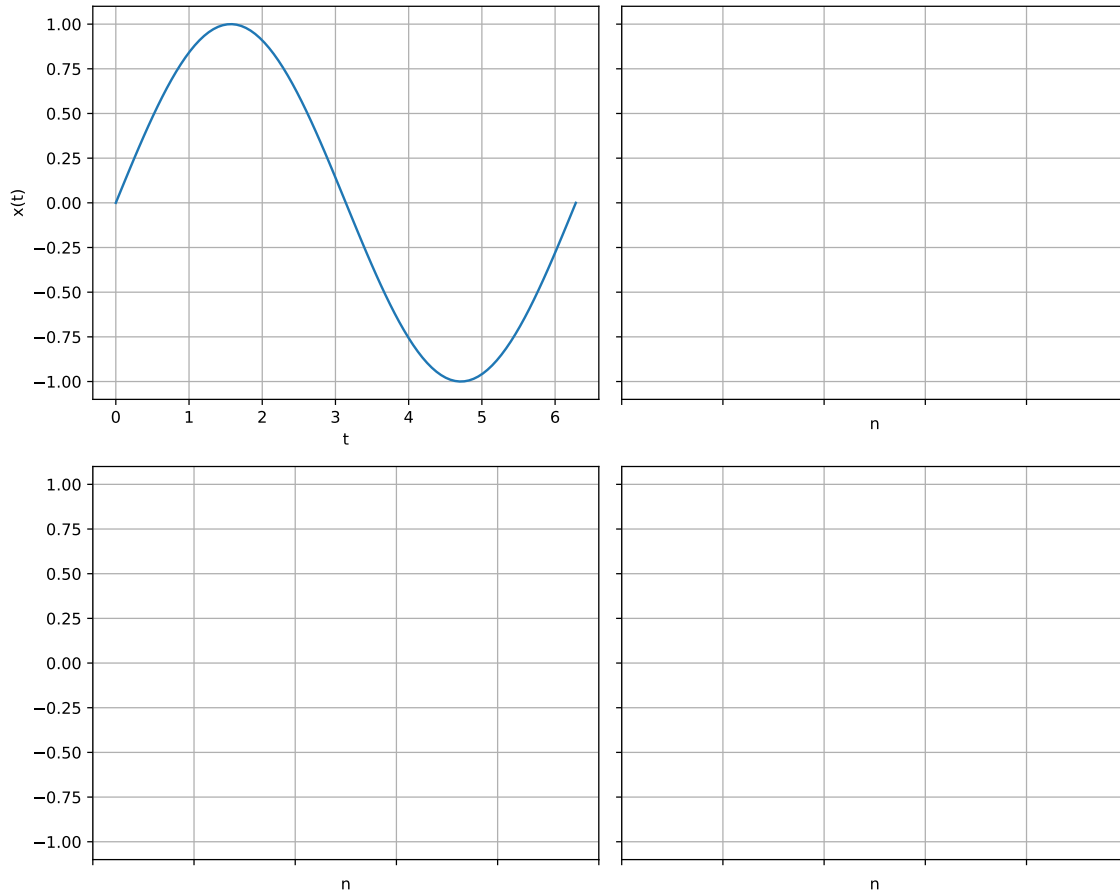
Suggested Oppenheim problems: 1.4, 1.5, 1.15, 1.21a-d, 1.22a-d, 1.27 and 1.28 (ignore “linear” and “time invariant” this week), 1.30a-c,e-g,k-n.

1 Continuous time vs. discrete time

In class we expressed two types of signals as mathematical functions of an independent variable (usually, but not always, time). In **continuous time**, a signal $x(t)$ is defined for every real number t . In contrast, in **discrete time**, a signal $x[n]$ is defined only for integer values of the independent variable, denoted n .

Someone asked a great question about whether n is strictly an integer. This is true, however, n may be an integer stand-in for an arbitrary real number. For example, imagine a signal comprised of the frames from 5 seconds of a movie filmed at 30 frames per second. While the “real time” of each frame is counted in increments of $1/30$ s, we can represent this as a discrete signal where n ranges from $1 - 150$.

1. Discuss with a neighbour whether the following signals are continuous-time or discrete-time. Are the answers clear cut, or ambiguous?
 - (a) the selling price of a stock
 - (b) the brightness of a fluorescent light
 - (c) the ambient air temperature outside during a flight
 - (d) your heart rate
2. Using the grids below, “discretize” (sample) the continuous signal $x(t) = \sin(t)$ for the three sets of time values. Convert time t to values of n and write them on the x axis (note you need not use the provided grid lines). Then, describe the qualitative differences between the three new signals. Can you tell they are all discretized versions of the same continuous signal?
 - (a) $x[n] = \sin(n)$ for n corresponding to $t = 0, 1, 2, 3, \dots, 6$
 - (b) $x[n] = \sin(n)$ for n corresponding to $t = 0.5, 1, 1.5, 2, \dots, 6$
 - (c) $x[n] = \sin(n)$ for n corresponding to $t = 0, \pi/2, \pi, 3\pi/2, 2\pi$



2 Transformations of independent variables

We applied three different transformations to the independent variable of both discrete and continuous time signals. In continuous time, we can compose the following transformations to send a signal $x(t) \rightarrow x(\alpha t + \beta)$

- Time shift: $x(t) \rightarrow x(t - t_0)$
- Time scaling: $x(t) \rightarrow x(at)$
- Time reversal: $x(t) \rightarrow x(-t)$

In discrete time, similar operations apply, but care is required due to the integer nature of n :

- Time shift: $x[n] \rightarrow x[n - n_0]$
- Time scaling: $x[n] \rightarrow x[a * n]$
- Time reversal: $x[n] \rightarrow x[-n]$

A systematic choice for order of operations is “*shift, scale, reverse*”: shift by β , scale by $|\alpha|$, then reverse if $\alpha < 0$. It’s possible to scale / reverse first and then shift, but you will need to factor out and correct apply the time shift. Practice this method on the signals in Figure 1 below.

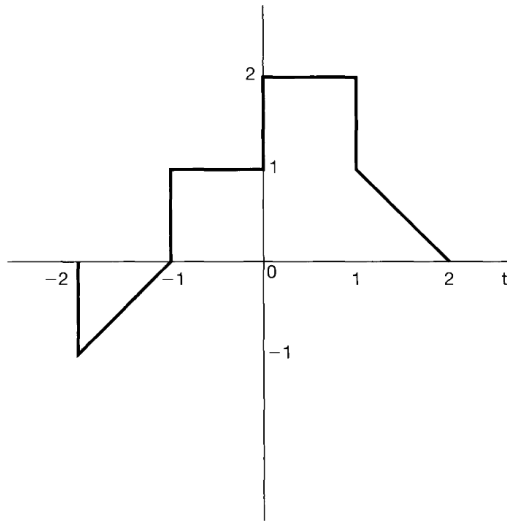


Figure P1.21

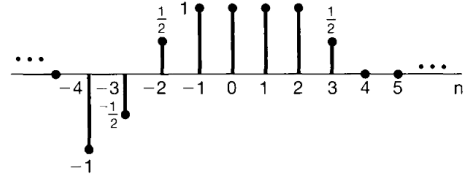


Figure P1.22

Figure 1: Image credit: Figures P1.21 and 1.22 from Oppenheim (p. 60).

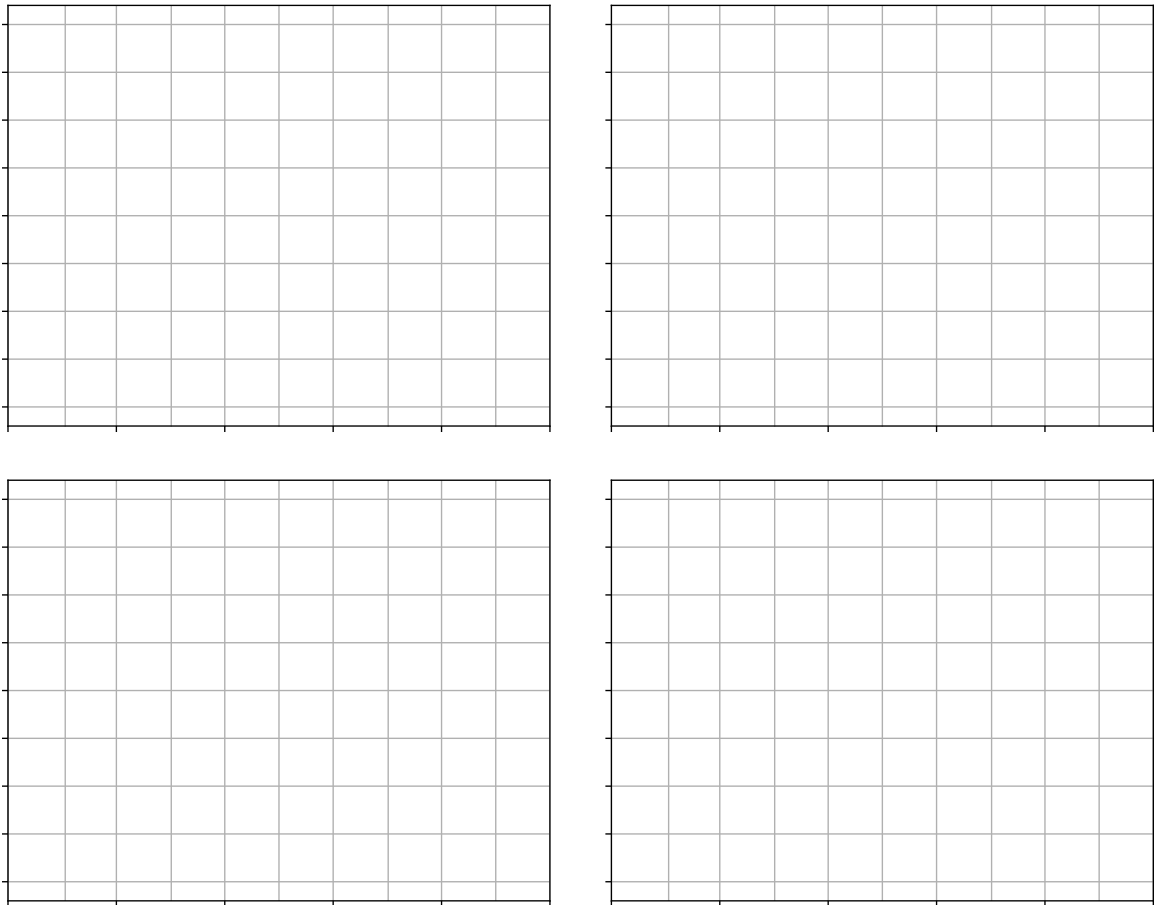


Figure 2: Transform the independent variable of the CT signal in Figure 1 to obtain the output signal $x(-0.5t - 1)$. Multiple grids are provided so you can work through the intermediate steps.

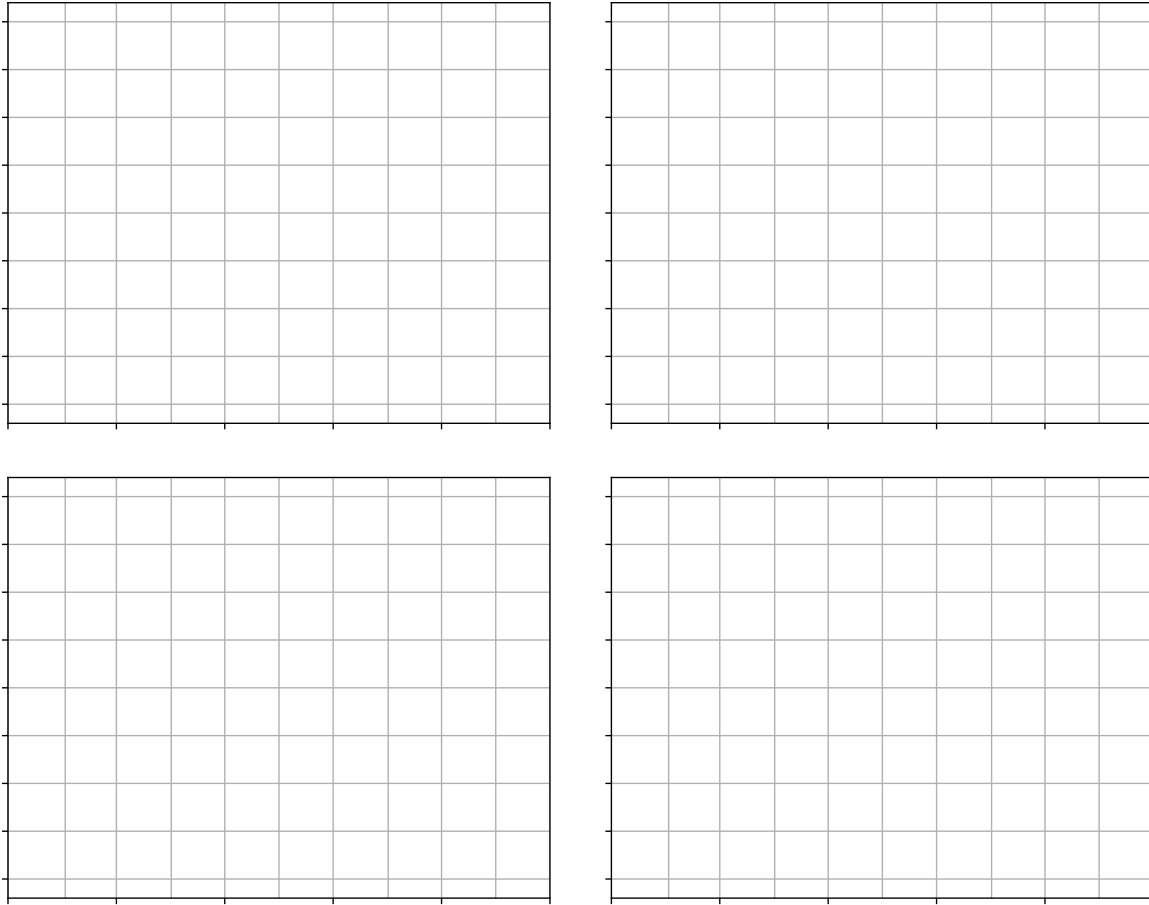


Figure 3: Transform the independent variable of the DT signal in Figure 1 to obtain the output signal $x[2n - 3]$. Multiple grids are provided so you can work through the intermediate steps.

3 System properties

In class we looked at four different system properties: memory(less), causality, invertibility, and stability.

3.1 Memory and causality

A system is **memoryless** if the system output at every time depends on the input signal at only that same time. A system is **causal** if the output at *any time* depends only on the input signal at the **present time or in the past**. Discuss with a neighbour the following systems are memoryless or causal, where $x(t)$ ($x[n]$) is the system input and $y(t)$ ($y[n]$) is the output.

- (a) $y(t) = \cos(3t)x(t)$
- (b) $y[n] = x[n] + x[n + 1]$
- (c) $y[n] = (n + 1)(x[n] - x[n - 1])$

3.2 Invertibility

Oppenheim provides two definitions for system invertibility:

1. “A system is said to be *invertible* if distinct inputs lead to distinct outputs”

2. “If a system is invertible, then an *inverse system* exists that, when cascaded with the original system, yields an output $w[n]$ equal to the input $x[n]$ to the first system.”

One approach to determine if a system is invertible is to construct an inverse system. Another, to prove a system is *not* invertible, is to find a counterexample, i.e., two input **signals** that lead to the same output signal (note: not input values, entire input *signals*). Later in the course, we will learn more formal tools for constructing inverse systems. For now, try to determine whether the following systems (examples from Oppenheim problem 1.30) are invertible.

(a) $y(t) = \cos(x(t))$

(b) $y[n] = x[4n]$

(c) $y[n] = \begin{cases} x[n/2], & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$

3.3 Stability

A system is **stable** if small changes in input do not cause the output to diverge. In other words, bounded inputs should lead to bounded outputs. Consider the systems below and determine whether they are stable. Assume the input signals have a minimum value of $-B$ and maximum value of B , for finite B . Note that later in the course, we'll also learn formal tools for analyzing stability of systems.

(a) $y(t) = x(t - 2) + x(2 - t)$

(b) $y[n] = y[n - 1] \cdot x[n]$

(c) $y(t) = x(t - 1) \cdot x(t)$

(d) $y[n] = n^{10}x[n]$